

INNOVA JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
in preparation for General Certificate of Education Advanced Level
Higher 2

CANDIDATE
NAME

ANSWERS

CLASS

INDEX NUMBER

BIOLOGY

9648/02

Paper 2 Core Paper

29 August 2017

2 hours

Additional Materials: Answer Paper
Cover Page

READ THESE INSTRUCTIONS FIRST

Write your name, class and index number on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use an HB pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the question paper.

Section B

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate. You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	11
2	10
3	12
4	12
5	12
6	11
7	12
Section B	
8 OR 9	20
Total	100

This document consists of **19** printed pages and **1** blank page.



Section A
Answer **all** questions.

- 1 Fig. 1.1 shows the mode of action of sodium-potassium pump found on the cell surface membranes of sweat glands.

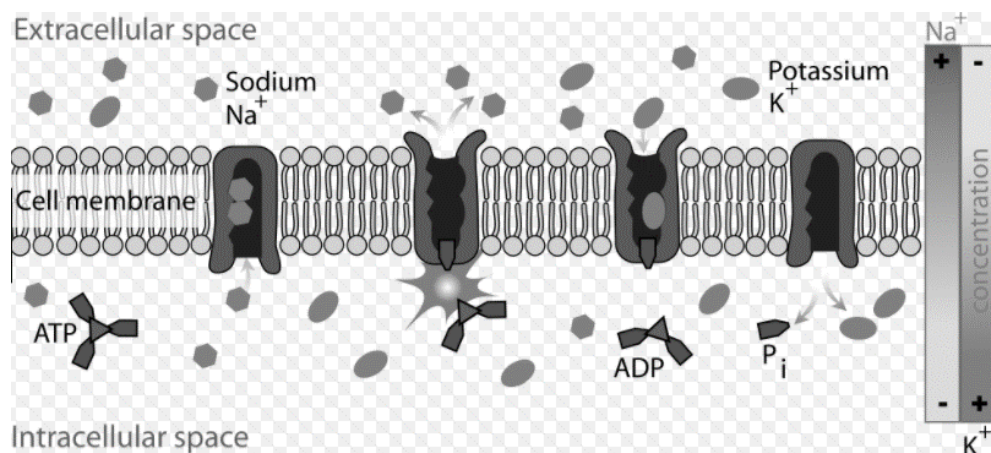


Fig. 1.1

- (a) Explain why the cell surface membrane is referred to as the *fluid mosaic model*.
1. fluid: phospholipid (& prot) are capable of lateral (&rotational) movmt in mem / within the monolayer

due to weak hydrophobic interxns

- 2. mosaic: prots of diff size & shape randomly distributed in bilayer**

OR presence of diff types of phospholipids in monolayers

mem is asymmetrical

[2]

- (b) Explain why ions can only move across cell surface membranes via transmembrane proteins.

- 1. ions are charged particles ® charged molecules**

unable to pass thru hydrophobic core (due to presence of FA tails) that repels ions

- 2. transmem prot provide a hydrophilic channel for ions to pass thru**

[2]

The sodium-potassium pump, also known as sodium-potassium ATPase, is able to catalyse the hydrolysis of ATP molecules to ADP and P_i . This reaction is coupled to the movement of ions across the membranes.

Movement of ions is based on the initial binding of Na^+ ions to the pump. This stimulates phosphorylation of the pump which causes a change in its conformation, leading to ion movement.

- (c) With reference to Fig. 1.1,
 (i) identify the type of transport mechanism that enables sodium ions to leave the cell;

active transport

[1]

- (ii) explain how the pump is also able to catalyse the hydrolysis of ATP molecules leading to the movement of ions across the membranes.

1. ATPase w active site compl to ATP

binds ATP → ES cplx

2. w catalytic residues that catalyzed hydrolysis

of phosphate bond resulting in ATP → ADP + Pi

3. energy released cause a conformational Δ in transmem prot

resulting in release of (3) Na⁺ to ECF

4. binding of K⁺ cause conformational Δ

leading to (2) K⁺ ions to entering cell

[3]

Ouabain is a cardiac glycoside that is able to affect the sodium-potassium pump. In an experiment to determine the effect of ouabain on the pump, sweat glands from rhesus monkeys were taken and treated to measure ATP hydrolysis. For each of the treatments, 5 sweat gland units were placed in a mixture of buffer and ATP.

Treatment 1: Sweat glands of rhesus monkey with Na⁺ and K⁺ added.

Treatment 2: Sweat glands of rhesus monkey with Na⁺, K⁺ and ouabain added.

Treatment 3: Sweat glands of rhesus monkey without additional Na⁺ and K⁺.

Table 1 shows the data collected at the end of the 1 hour experiment.

Table 1

Treatment	Number of moles of ATP hydrolysed / g ⁻¹
1	0.58
2	0.37
3	0.38

- (d) Explain the effects of ouabain on ATP hydrolysis.

1. less no. of moles of ATP hydrolysed with ouabain added (Trmt 3) c.f. to without (Trmt 1)

[QV] 0.37 g⁻¹ vs 0.58 g⁻¹

2. ouabain acts as inhibitor of Na⁺/K⁺ ATPase

occupying active site of enz to ⊗ ATP from binding

[2]

- (e) Suggest a reason for the results of Treatment 3.

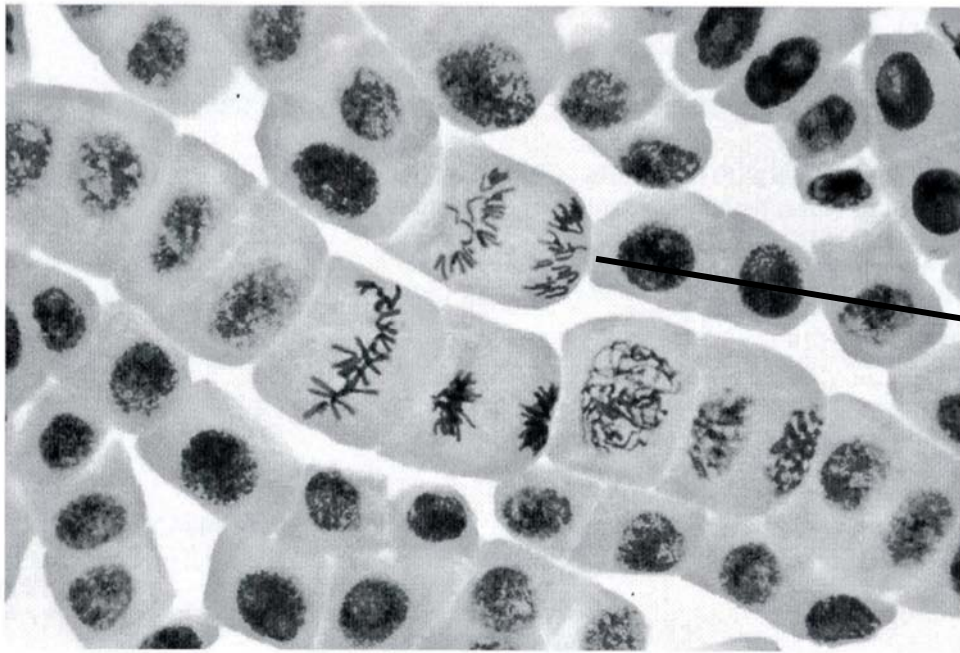
1. cells have existing Na⁺ & K⁺ thus ATPase is able to hydrolyse some ATP

but only 0.37 g⁻¹ hydrolysed as Na⁺/K⁺ is depleted

[1]

[Total: 11]

2 Fig 2.1 is a photomicrograph of plants cells, with some undergoing mitosis.



anaphase

Fig. 2.1

- (a) On Fig 2.1, use labels and label lines to indicate **one** cell in anaphase stage of mitosis. [1]
- (b) The longest stage of the mitotic cell cycle, interphase, is divided into three phases, **G1, S and G2**.

- (i) Describe what happens in the G1 phase.

1. syn of organelles e.g. mito

2. syn of prot

3. store en. (any MP 1 – 3 for 1m)

4. increase in cell size (↑ cytoplasmic vol)

5. nucleolus actively syn rRNA → formation of ribo

[3]

- (ii) There are various checkpoints in the mitotic cell cycle. One of them is present in the G1 phase called the G1 checkpoint.

Describe the function of G1 checkpoint.

1. checks for presence of growth factors

cell enters G1 arrest if GF is absent

2. checks for DNA damage

trigger DNA repair or apoptosis (if beyond repair)

[2]

- (iii) When cell cycle checkpoints are defective, cancer could arise.

With reference to two named genes, outline the development of cancer.

1. GoF mutation of proto-oncogene e.g. Ras → oncogene

LoF mutation of TSG e.g. p53

2. activation of telomerase gene

active telomerase allows mutant cells have unlimited replicative potential

3. mutation allows cell to evade apoptosis

leading to uncontrolled cell division

4. activation of genes involved in angiogenesis → proliferation of blood vessels to tumour cells

mutation of genes involved in cell-cell adhesion → metastasis [4]

[Total: 10]

- 3** Fig. 3.1 shows a scanning electron micrograph of *Escherichia coli* under mass attack by numerous phage-T4 virions.

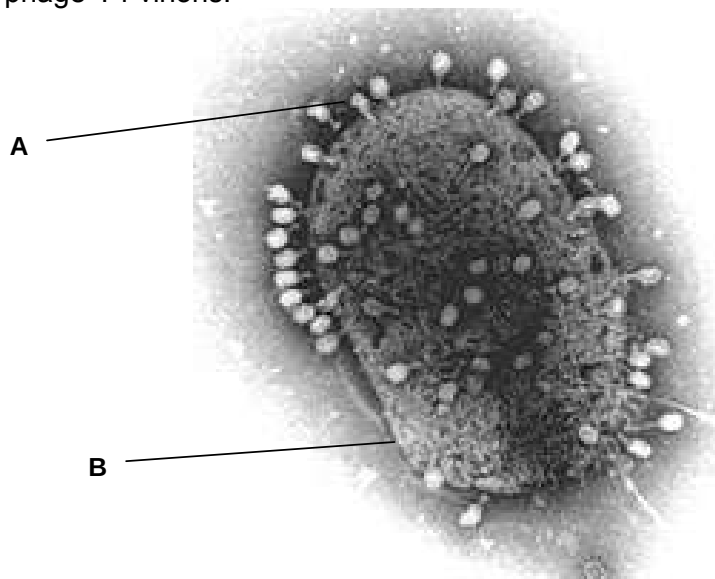


Fig. 3.1

- (a)** Identify and briefly describe the function of the structures labelled **A** and **B** in Fig. 3.1.

1. A: capsid / nucleocapsid

Fn: protect viral nucleic acids (from digestion by nucleases)

2. B: bacteria cell wall

Fn: maintains shape of bacterium / protect from mechanical damage [2]

- (b) Describe how T4 phage attaches and penetrates into *Escherichia coli*.
- 1. long tail fibres bind to surface receptor on bacterium via weak chemical bonds**

.....

- 2. base plate pins form strong & irreversible bond**

.....

- 3. sheath of tail contracts**

.....
**hollow core puncture bacterial cell wall, injecting phage DNA into
 bacteria cell cytoplasm**

[3]

Lambda (λ) phage infections result in the lysis of the bacterial host cells which shows up as a plaque or a zone of clearing on an agar plate containing a dense growth of bacteria.

This observation can be used to quantify the number of infectious bacteriophage particles through a plaque assay. 'Plaque Forming Unit (PFU)' gives the measurement of the number of infectious viral particles present during the experiment.

Fig. 3.2 below shows the changes in PFU caused by lambda (λ) phages infecting *E.coli* cells upon irradiation by UV.

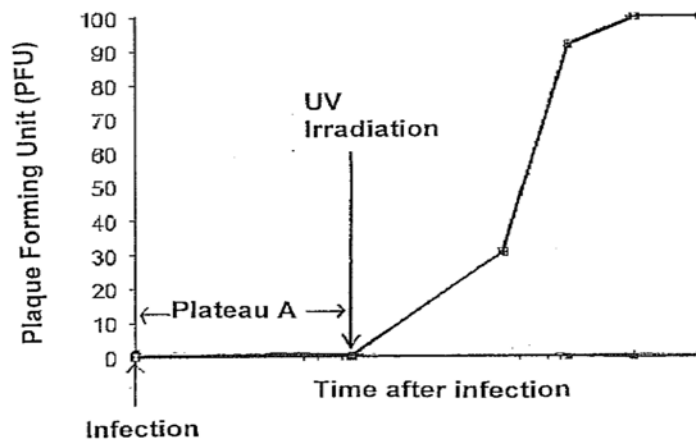


Fig. 3.2

- (d) State the difference in the type of life cycle between T4 and λ phage.

- 1. T4 phage undergoes lytic life cycle**

.....
 λ undergoes lysogenic (and lytic life cycle)

[1]

- (e) With reference to Fig. 3.2,

(i) suggest an explanation for the flat part of the graph, labelled 'Plateau A'.

- 1. at A, from start of infection till UV irradiation PFU remains constant at 0**

.....
virus is in latent phase

- 2. viral genome integrated into host genome to form prophage**

.....
prophage is transcriptionally silent (suppressed by repressor prot)

- 3. viral DNA is replicated as host cell replicates (by binary fission)**

.....
no host cell lysis to release phage particles

[3]

- (ii) Account for the rise in PFU shortly after UV irradiation.

1. **PFU ↑ from 0 to 30 / 100**

as phage particles are released from host cells by lysis of bacterial cell wall

2. **prophage is spontaneously induced (by UV irradiation)**

to enter lytic cycle

3. **prophage excised from bacterial chromosome**

replicate → viral genome, transcribe & translate viral DNA → viral prot (using host cell machinery)

viral prot are assembled with viral DNA into new phage to be released

[3]

[Total: 12]

- 4 The flowers of a species of *Primula* may have a yellow 'eye' in the centre of the flower or be 'eyeless'. The 'eye' may be small or large, as shown in Fig. 4.1.

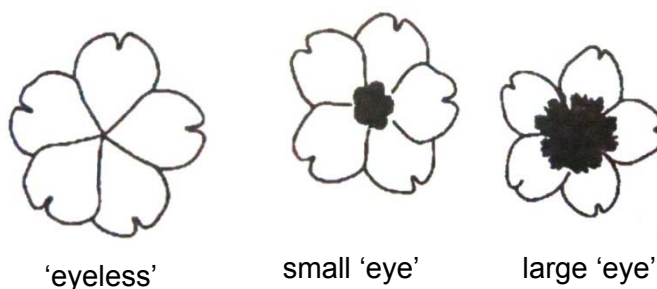


Fig. 4.1

These phenotypes are produced by three alleles of a gene. Allele A^S suppresses yellow pigment production in the 'eye' giving 'eyeless' flowers. Allele A^S is dominant to allele A^R , which restricts the size of the eye, resulting in flowers with small 'eyes'. Both A^S and A^R are dominant to a .

- (a) (i) Explain what is meant by dominant allele.

1. **an alternative form of a gene**

that is expressed in phenotype regardless of the allelic counterpart / allele at other homolog

[1]

- (ii) List all the genotypes that give rise to each type of flower.

Phenotypes	Genotypes
'eyeless' flowers	$A^S A^S, A^S A^R, A^S a;$
"small-eyed" flowers	$A^R A^R, A^R a;$
'large-eyed' flowers	$aa;$

[3]

- (iii) Suggest a mechanism by which allele A^S suppresses yellow pigment production.

1. A^S codes for a prot that inhibits enz involved in syn of yellow pigment OR

2. codes for enz that breaks down yellow pigment

3. thus 1 copy of A^S allele is sufficient to prod prot / enz that results in no yellow pigment

4. @ A^S is epistatic over gene coding for yellow pigment

[2]

® A^S is epistatic over allele A^R and a

- (b) A plant with the genotype $A^S a$ was crossed with one with the genotype $A^R a$.

Draw a genetic diagram to show the gametes and the genotypes and phenotypes of the offspring of this cross. Give the expected ratio of offspring phenotypes.

Parental phenotype	eyeless male		x	small eyed female											
Parental genotype	$A^S a$		x	$A^R a$											
Gametes	A^S a			A^R a											
Random fertilization	<table><tr><td></td><td> A^S </td><td> a </td></tr><tr><td> A^R </td><td>$A^S A^R$</td><td>$A^R a$</td></tr><tr><td> a </td><td>$A^S a$</td><td>aa</td></tr></table>							A^S	a	A^R	$A^S A^R$	$A^R a$	a	$A^S a$	aa
	A^S	a													
A^R	$A^S A^R$	$A^R a$													
a	$A^S a$	aa													
offspring genotype	$1 A^S A^R$: $1 A^S a$: $1 A^R a$: $1 aa$														
offspring phenotype	$2 \text{ eyeless} :$ $1 \text{ small eyed} :$ 1 large eyed [2]														

- (c) In such a cross between plants with genotypes $A^S a$ and $A^R a$, the following numbers of phenotypes of offspring were recorded.

phenotype of flower	number of offspring
'eyeless' flowers	85
"small-eyed" flowers	59
'large-eyed' flowers	56

- (i) Use the chi-squared (χ^2) test and the table of probabilities shown in the **Table 4** to find the probability of the results of this cross departing significantly by chance from expectation. Show your working.

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

O = observed result

E = expected result

$v = n - 1$

Table 4

degrees of freedom	probability P				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.42	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

	<i>O</i>	<i>E</i>	<i>O - E</i>	$(O - E)^2$	$(O - E)^2 / E$
<i>Eyeless</i>	85	100	-15	225	2.25
<i>small eyed</i>	59	50	9	81	1.62
<i>large eyed</i>	56	50	6	36	0.72
<i>Total</i>	200	200	0		$\chi^2 = 4.59$

1. correct expected values
correct χ^2 value
2. $p > 0.10$

[2]

(ii) State the conclusion that could be drawn from the probability found in c(i).

1. calculated χ^2 value is < critical value of 5.99

at 5% level of significance, $df = 2$

2. deviation b/w *O* & *E* is statistically insignificant

diff is due to chance (ratio observed is 9:3:3:1)

[2]

[Total: 12]

- 5 Fig 5.1 shows a ligand binding to the G Protein-coupled receptor (GPCR) which is embedded on the cell surface membrane.

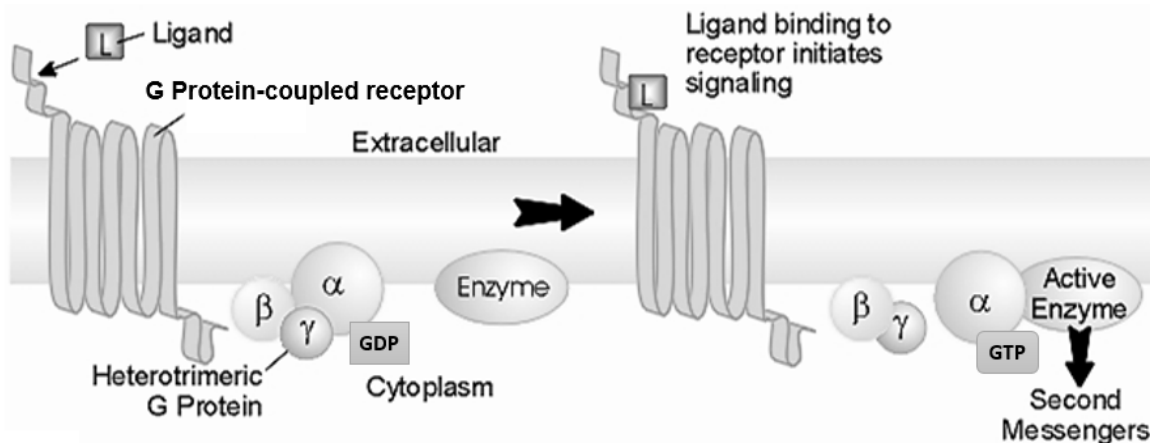


Fig. 5.1

- (a) (i) Identify the ligand in Fig. 5.1.

glucagon

[1]

- (ii) Explain why ligand mentioned in a(i) unable to pass through the cell surface membrane.

1. *glucagon is a peptide hormone / prot*

which is a large macromolecule

2. *is amphipathic*

repelled by hydrophobic core of CSM

3. *no transport protein (carrier / channel) large enough to facilitate its transport across CSM*

[2]

- (b) With reference to Fig 5.1, describe what happens when the ligand binds to the GPCR.

1. *ligand binds to ligand-binding site of GPCR via complementary fit*

receptor undergoes conformational Δ , activating it (via ligand receptor interxn)

2. *GDP on $G\alpha$ is displaced by GTP, activating it*

$G\alpha$ dissociate from $G\beta\gamma$ dimer & translocate along mem to activate enz (adenylyl cyclase)

3. *activates enz to syn 2nd msgers*

resulting signal transduction (causing cellular response)

[3]

- (c) With a named example, define "second messengers".

1. *are small, non-prot, water soluble molecules*

e.g. cAMP

2. *that relays signals from cell surface receptors to target molecules in the cell*

by activating relay prot (e.g. PK)

[2]

- (d) Explain how intracellular signal is terminated when ligand is released from the receptor.

1. *GTPase in $G\alpha$ hydrolyse bound GTP $\rightarrow$ GDP*

$G\alpha$ reassociate with $G\beta\gamma$

2. *phosphodiesterase cleaves cAMP $\rightarrow$ AMP*

$\downarrow$ 2nd msger

[2]

- (e) One of the side effects of a particular drug includes non-responsiveness of GPCR to ligands.

Suggest how the drug could have caused such non-responsiveness of GPCR to

ligands.

1. **drug has a similar 3D config as ligand / glucagon**

(act as competitive inhibitor) competes with ligand for ligand binding site on GPCR

2. **binds irreversibly / permanently**

⊗ ligand from binding to GPCR to initiate signal transduction OR

3. **non-competitive inhibitor**

binds to allosteric site on GPCR

4. **cause conformational Δ to GPCR ligand-binding site**

⊗ longer complementary to ligand

[2]

[Total: 12]

6 ATP synthase is an enzyme that synthesizes ATP from ADP and inorganic phosphate (P_i).

Fig. 6.1 shows the structure of ATP synthase. It is found in both the chloroplast and the mitochondrion.

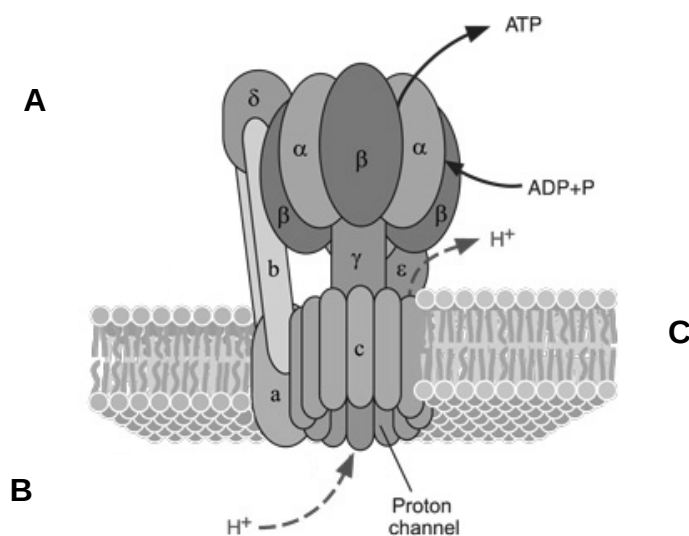


Fig. 6.1

(a) In Table 6.1, indicate the regions A, B and C that are found in mitochondrion and chloroplast respectively.

Table 6.1

organelle	A	B	C
chloroplast	stroma	thylakoid space	thylakoid mem
mitochondrion	matrix	intermem space	inner mito mem

[2]

In both organelles, a proton gradient exists across the membrane where ATP synthase is located. The synthesis of ATP by ATP synthase is coupled with the diffusion of H^+ across the proton channel down its concentration gradient.

(b) Describe how this proton gradient is generated in the chloroplast.

1. photon of light absorbed photoactivates e^- from chl a in rxn ctr of PSII

e^- passed down ETC

2. en released from e^- trspt

used to pump H^+ from stroma into thylakoid space

3. thylakoid memb impermeable to H^+

due to repulsion by hydrophobic core $\rightarrow$ creating proton grad / PMF across thylakoid mem

4. photolysis of H_2O by water splitting enz in thylakoid lumen

$H_2O \rightarrow 2H^+ + \frac{1}{2} O_2 + e^-$

[3]

The blue dye DCPIP can be converted to colourless reduced DCPIP by gaining electrons.

A suspension of chloroplasts was made by grinding fresh leaves in buffer solution and centrifuging the mixture. Tubes were then prepared and treated as shown in Table 6.2.

Table 6.2

Tubes	Content	Treatment	Colour	
			Start	After 20 mins
A	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Illuminated strongly	Blue-green	Green
B	1 cm ³ buffer solution 5cm ³ DCPIP	Illuminated strongly	Blue	Blue
C	1 cm ³ chloroplast suspension 5cm ³ DCPIP	Left in the dark	Blue-green	Blue-green

(c) With reference to Table 6.2,

(i) name a possible source of electrons that reduces DCPIP;

PSII / chl a / PSI / water

[1]

(ii) explain this observation using your knowledge on light-dependent reactions of photosynthesis.

1. DCPIP is reduced from blue to colourless in the presence of chlplst & light

as photon of light photoexcites e^- s in rxn ctr passed down ETC in thylakoid mem

2. e- deficit in PSII**is replaced by e- from photolysis of H₂O****3. DCPIP act as final e- acceptor in non-cyclic photophosphorylation****in place of NADP+**

[3]

(d) Paraquat is a weedkiller. It binds with electrons of PS I.

Suggest how paraquat results in the death of a plant.

1. e⁻ not trspted down ETC**⊗ cyclic & non-cyclic photophosphorylation****2. no ATP & NADPH for conversion of GP to TP (reduction phase) in Calvin cycle****no glucose (for respiration)**

[2]

[Total: 11]

- 7 The marine threespine sticklebacks, *Gasterosteus aculeatus* is a freshwater fish living in the lakes of British Columbia, Canada as shown in Fig. 7.1.

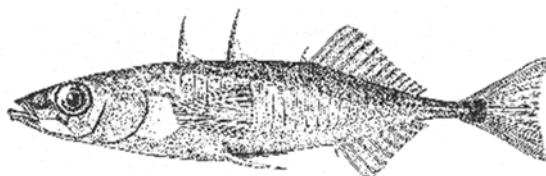


Fig. 7.1

In order to investigate the process of speciation in these populations, three small lakes were studied. Each lake contained two varieties of stickleback: a large, bottom-dwelling variety that fed on invertebrates near the shore and a small, plankton-eating variety that lived in the open water. The probability of breeding between pairs of individuals was measured under laboratory conditions in the following breeding combinations:

I	different varieties from the same lake
II	different varieties from different lakes
III	same variety from different lakes
IV	same variety from the same lake

The data are summarized in Fig. 7.2 below.

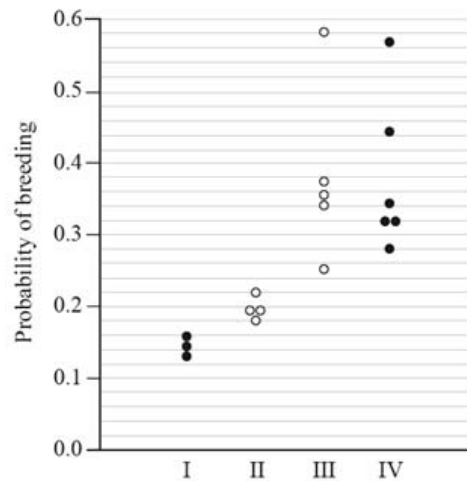


Fig. 7.2

(a) With reference to Fig. 7.2,

- (i) identify the highest and lowest probabilities of breeding for individuals of the same variety

highest probability is 0.58 & lowest probability is 0.25

[1]

- (ii) describe the differences in probability of breeding between individuals from different lake.

1. individuals from diff variety have lower breeding probability

c.f. to individuals from same variety

2. $0.18 < p < 0.22$ c.f. $0.25 < p < 0.58$

[2]

- (iii) describe the evidence that speciation is taking place in these populations and explain the type of speciation as shown in Fig. 7.2.

1. prob of diff varieties interbreeding is low even if they are in the same lake

$0.13 < p < 0.16$

2. accumulation of genetic diff btw varieties

leading to behavioural / physiological isolation

3. resulting in (pre- / post-zygotic) reproductive barriers

sympatric speciation may be occurring

[3]

- (iv) explain why all the individuals are still considered the same species.

1. based on BSC

both varieties can still interbreed to produce viable, fertile offspring

2. probability of breeding btw diff variety

from same lake is $0.13 < p < 0.16$

from diff lake is $0.18 < p < 0.22$

[2]

- (b) The freshwater lakes also contain many different types of parasites that infect the different varieties of marine threespine sticklebacks. Explain why these parasites help to speed up speciation of the marine threespine sticklebacks.

1. diff type of parasites infecting specific varieties of sticklebacks

acts as a selection pressure

2. sticklebacks with diff alleles which confer resistance to specific parasite are selected for

➡ *higher survival rate & repro success*

3. passing down these alleles to their offspring

➡ Δ *in allelic frequency in the gene pools of the diff varieties*

4. diff gene pools accumulate diff alleles resistance to diff parasite

➡ *genetic divergence which leads to reproductive isolation speeding up speciation* [4]

[Total: 12]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in section **(a)**, **(b)** etc., as indicated in the question.

8 (a) Compare prokaryotic and eukaryotic genomes. [8]

- 1. Prokaryotic chromosome is circular while eukaryotic chromosome is linear;;**
- 2. Prokaryotic chromosome has one origin of replication while the eukaryotic chromosome has multiple origins of replication;;**
- 3. Prokaryotic DNA is associated with histone-like protein while eukaryotic DNA associated is with histone octamer and H1 protein to form nucleosomes;;**
- 4. Prokaryotic chromosome has no centromeres & telomeres while eukaryotic chromosome has centromere & telomeres made up of tandem repeats;;**
- 5. Prokaryotic chromosome has little / no introns while eukaryotic chromosome has exons & introns;;**
- 6. In prokaryotes, there is presence of operons as genes coding for proteins in the same metabolic pathway are under the control of a single promoter while in eukaryotic genome there is no operon i.e every gene is under the control of a single promoter;;**
- 7. In prokaryotes, gene expression can be controlled by at transcriptional level by operator while in eukaryotes, gene expression is regulated at multiple levels (eg. chromosomal modification/transcriptional/post-transcriptional/translational/post-translational);;**
- 8. In prokaryotes the genes are in a single copy while in eukaryotes there are 2 copies of a gene in a diploid state;;**
- 9. In prokaryotes there are lesser genes while in eukaryotes there are more genes present;;**

- (b) Explain how the expression of a gene expression is decreased by chromatin modification. [6]

1. ***Histone deacetylation involves removing an acetyl group from lysine residues at the N-terminal of histone tails catalyzed by histone deacetylase;;***
2. ***increases positive charge on the histone increases histone's affinity for negatively charged DNA;;***
3. ***makes the DNA to coil more tightly around the histone to result in a compact conformation of the DNA;;***
4. ***DNA methylation involves covalent addition of methyl group To the cytosine residues in the CpG island;;***
5. ***to form methyl-cytosine catalysed by methyl transferase;;***
6. ***results in decreased accessibility of the promoter region to RNA polymerase and transcription factors thus decreasing transcription rate;;***

- (c) Explain why processing of pre-mRNA molecules is necessary for regulation of gene expression. [6]

1. ***pre-mRNA transcribed from DNA contains introns which are non-coding; spliceosome splices the intron and join the exons;;***
2. ***this prevents extra amino acids from being incorporated during translation which will form non-functional protein;;***
3. ***allows alternative splicing removes all introns and join different exons;;***
4. ***results in different forms of protein to be produced from 1 type of pre-mRNA to regulate expression of the gene / amount of protein produced;;***
5. ***5' end of the mRNA is capped by the addition of a methylated guanine forming a 5' cap;;***
6. ***adenine residues are attached to the 3' end of the mRNA forming a 3' poly(A) tail ;;***
7. ***5' cap and 3' poly(A) tail facilitates the export of mRNA to the cytoplasm/ prevent enzymatic degradation which controls mRNA stability;;***

[Total: 20]

- 9 (a) Using *lac* operon as an example, describe how prokaryotes such as *Escherichia coli* is able to respond to changes in the environment.
[8]

When lactose is present (negative control)

1. *lactose enters thru lactose permease on cell memb
isomerizes to allolactose by beta-galactosidase;;*
2. *allolactose binds to repressor prot
causing repressor to change its conformation & becomes inactive;;*
3. *DNA binding site of repressor is no longer complementary to operator &
no longer binds
RNA pol can bind to promoter & transcribe structural genes;;*

When lactose is absent (negative control)

4. *lac repressor encoded by lac I gene produced is active
lac repressor DNA binding site binds to operator site;;*
5. *RNA pol unable to transcribe structural genes as promoter site is blocked*

When both glucose & lactose are present (positive control)

6. *cAMP levels will be low & CAP remains inactive
no formation of CAP-cAMP complex;;*
7. *CAP-cAMP complex does not bind to CAP binding site
RNA pol affinity to promoter is low & rate of transcription occurs
minimally;;*

When glucose absent/low levels & lactose present (positive control)

8. *cAMP produced from ATP, cAMP binds to inactive CAP to form CAP-
cAMP complex
complex binds to binding site on promoter;;*
9. *strengthens binding of RNA pol to promoter enhancing transcription
rate;;*

(b) Contrast binary fission and mitosis.

[6]

<i>Basis of comparison</i>	<i>Binary fission</i>	<i>Mitosis</i>
<i>type of cells that process occurs in</i>	<i>1. occurs in prokaryotes</i>	<i>1. occurs in eukaryotes</i>
<i>location of cell that process occurs in</i>	<i>2. occurs in nucleoid / cytoplasmic region of prokaryotic cell</i>	<i>2. occurs in nucleus of eukaryotic cell</i>
<i>type of DNA involved</i>	<i>3. occurs on circular double-stranded DNA</i>	<i>3. occurs on linear double stranded DNA.</i>
<i>no. of chromosomes involved</i>	<i>4. only 1 chromosome involved</i>	<i>4. more than 1 chromosome involved</i>
<i>separation of daughter cells</i>	<i>5. division of parental cell to give rise to 2 daughter cells</i>	<i>5. no division of parental cell to give rise to 2 daughter cells, occurs during cytokinesis</i>
<i>type of division</i>	<i>6. involves cell division</i>	<i>6. involves nuclear division</i>
<i>origin of replication</i>	<i>7. DNA replication occurs at 1 point of origin</i>	<i>7. DNA occurs at multiple points of genome</i>
<i>utilisation of spindle fibres</i>	<i>8. no spindle fibre formation</i>	<i>8. spindle fibre formation occurs</i>
<i>presence/ absence of end replication problem</i>	<i>9. no end replication problem</i>	<i>9. end replication problem occurs</i>

(c) Explain the significance of genetic exchange in bacteria. [6]

1. *via transformation, a recipient cell takes up small fragments of naked, foreign DNA from surrounding env incorporated into recipient cell via recombination;;*
2. *via transduction, bacteriophages carry bacterial genes from first host cell & infect into second host cell homologous recombination of donor DNA fragment takes place with a homologous section of recipient cell's chromosome;;*
3. *via conjugation where direct contact / cytoplasmic bridge b/w donor & recipient cells results in transfer of F factor/ plasmid from F⁺ to F⁻ cell;;*
4. *above processes allow for genetic recombination, resulting in genetic variation in bacterial populations;;*
5. *essential for natural selection, selective advantage / differential survival & reproduction / better adapted to environment;;*
6. *transfer of beneficial genes such as antibiotic resistance genes;;*

[Total: 20]

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